Sarthak Kumar Maurya

+917011445426 | sarthakmaurya04@gmail.com | linkedin.com/in/sarthax11 | github.com/sarthexe

Education

Guru Gobind Singh Indraprastha University — Delhi, India
B.Tech. in Artificial Intelligence and Machine Learning — *Nov 2022 – June 2026*

Experience

AI Engineer — Apr 2026 – Present
Neolytix — *Gurugram, Haryana*

- Shipped **NeoSutra**, a production **RAG assistant** using **FastAPI**, **pgvector**, and **Claude**, supporting **20 client knowledge bases** with **91.2% retrieval coverage**.
- Improved **RAG quality** through retrieval and prompt tuning, reducing hallucinations by **8%** and achieving **94% grounding** across responses.
- Built **SharePoint synchronization** with **Microsoft Graph**, automatically applying **360 changes** with a **0.8-minute median** detection-to-knowledge-base update time.
- Developing **NeoForge**, an **AWS AI gateway** using **LiteLLM** and **Amazon Bedrock** for PHI workloads, with per-tenant budgets, model controls, and **PII guardrails**.
- Engineered fail-closed **PHI de-identification** with **Comprehend Medical**, achieving **94.4% recall** and validating the system through **5 release gates** and **634 tests**.

AI/ML Intern — Jul 2025 – Apr 2026
Neolytix — *Gurugram, Haryana*

- Built a CPT Code Recommender Agent in **Microsoft Copilot Studio** achieving **87% recommendation** accuracy across 5,000 test queries for medical coding workflows.
- Developed an **MCP Server Inspector** that validated **15+ tools** across **40+ integration checks**, catching schema and invocation failures before agent deployment.

Freelance AI Engineer (Contract) — Jan 2026 – Apr 2026
Airtym — *Remote*

- Developed a real-time **AI voice platform** using **FastAPI**, WebSockets, **Deepgram**, **Groq**, and **ElevenLabs** for streamed conversations and interruption handling.
- Optimized a **pgvector-backed RAG pipeline**, reducing **retrieval p95 by 24.7%** (2.35s to 1.77s) across 50 matched queries and **post-transcript first-audio p95 by 12.2%** across 30 live-provider turns.
- Built **persona-aware retrieval** and **contact memory** using **Voyage AI**, PostgreSQL/pgvector, token-aware processing, and background workers.

Projects

Job Agent AI | *Python, LangGraph, FastAPI, Qdrant, PostgreSQL, Playwright*

- Built a **multi-agent job automation platform** with **LangGraph**, **FastAPI**, **Qdrant**, PostgreSQL, and **Playwright** for job discovery, resume matching, tailoring, and browser-based application workflows.

Tiny LLM | *Python, PyTorch*

- Implemented a **GPT-style language model from scratch** in **PyTorch** with token/positional embeddings, causal multi-head attention, transformer blocks, training, and autoregressive generation.

Technical Skills

Languages: Python, C, SQL, JavaScript,Rust
AI / GenAI: LLMs, RAG, AI Agents, LangChain, LangGraph, MCP, Tool Calling, Prompt Engineering
ML /Deep Learning: PyTorch, Transformers, Hugging Face, scikit-learn, XGBoost
Retrieval / Data: PostgreSQL, pgvector, Qdrant, FAISS, Embeddings, Semantic Search
Backend / Cloud: FastAPI, Docker, AWS, Amazon Bedrock, LiteLLM, LangSmith, Ragas, Git